The FST-RH Framework: A Gauge-theoretic Reformulation of the Riemann Hypothesis

Executive Summary and Consolidated Thesis*

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1 Overview: The FST-RH Trilogy

The FST-RH framework develops a gauge-theoretic perspective on the Riemann Hypothesis (RH), connecting thermodynamic formalism on primes to the Li coefficients via the completed zeta function $\xi(s)$. This trilogy explores the bridge between analytic number theory and operator-theoretic stability.

Part I: Prime Hub Graphs and the Vacuum Trace constructs the motivational bridge. It defines the *Prime Hub Graph* transfer operator and argues that the zeros of $\zeta(s)$ appear as its resonances. It introduces the Maximum Entropy Production Principle (MEPP) as a heuristic driver for critical-line symmetry.

Part II: Analytic Obstructions and the Limits of Monotonicity establishes the Bombieri–Lagarias decomposition $\lambda_n = \lambda_n^{(\infty)} + \lambda_n^{(\mathrm{fin})}$ and identifies the "Archimedean Dominance Paradox": the archimedean Gamma-block provides global convexity, while the arithmetic prime-sum part $F(\sigma)$ changes sign at $\sigma_0 \approx 1.14$. It shows that Gibbs-normalized prime-sum formulations capture only local structure near $\sigma = 1$ and cannot serve as a global proxy for the Li coefficients.

Part III: Spectral Analysis and Certificate Structure analyzes the spectral structure of the symmetrized arithmetic kernel, finding a stable negative inertia index of 2. It develops the S-procedure certificate framework and discusses the scope and limitations of the gauge-correction approach.

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2 Key Results and Honest Assessment

2.1 What is Rigorously Established

1. **Bombieri–Lagarias Decomposition** (Part II): $\lambda_n = \lambda_n^{(\infty)} + \lambda_n^{(\mathrm{fin})}$ via the completed xi function $\xi(s)$, with $\lambda_n^{(\mathrm{fin})}$ computed from $g(s) = (s-1)\zeta(s)$, which is holomorphic at $s = 1$ [1].

2. **Convergence of Prime-Sum Functionals** (Part II): $A(\sigma), C(\sigma), D(\sigma)$ converge absolutely at $\sigma = 1$. The target functional $F(\sigma) = AB - P(C + D)$ is positive for $\sigma \in (1, 1.14)$ and negative for $\sigma > 1.14$.

3. **Spectral Census** (Part III): The symmetrized arithmetic kernel has a stable negative inertia index of 2 for $N \geq 300$ primes and $\sigma \in (1, 5)$.

4. **Gauge Equivalence** (Part III): For each fixed σ, the S-procedure provides a tautological equivalence between $F(\sigma) \geq 0$ and the existence of an admissible PSD gauge.

2.2 What Does NOT Work

1. **Global Gauge Reduction**: The claim "RH $\Leftrightarrow$ $F(\sigma) \geq 0$ for all $\sigma > 1$" is false, since $F(\sigma) < 0$ for $\sigma > 1.14$ independently of RH. The Gibbs-normalized prime-sum formulation cannot serve as a global proxy for the Li coefficients.

2. **Gibbs Normalization**: The expectation $\mathbb{E}_{\mu_\sigma}[X_{n,\sigma}] = -A(\sigma)/P(\sigma)$ annihilates the finite-part contribution as $\sigma \to 1^+$ (the $1/P(\sigma)$ normalization kills the finite rest). The "Critical Identification" in the Gibbs framework converges to 0, not λ_n.

3. **Monotonicity Argument**: The true function $\frac{d}{ds}\log\xi(\sigma)$ is *increasing* (convex, $\frac{d^2}{ds^2}\log\xi > 0$), not decreasing. The FST observable $m_n(\sigma)$ is decreasing but has the wrong limit.

2.3 Open Path Forward

The core insight of the FST-RH program — that the Li coefficients decompose into competing archimedean and arithmetic contributions, with archimedean dominance ensuring positivity — remains valid as a structural observation. Numerical analysis reveals a three-regime structure:

- **Small n ($n \lesssim 8$):** $\lambda_n^{(\infty)} < 0$ but $\lambda_n^{(\mathrm{fin})} > 0$ compensates. Positivity is a delicate cancellation.
- **Transition ($n \approx 8$):** $\lambda_n^{(\infty)}$ crosses zero; both parts contribute positively.
- **Large n ($n \gg 10$):** $\lambda_n^{(\infty)} \sim \frac{n}{2}\log n$ dominates; positivity is automatic.

Under RH, $|1 - 1/\rho| = 1$ for all zeros, giving $\lambda_n = \sum_\rho [1 - \cos(n\theta_\rho)] \geq 0$. Without RH, off-line zeros produce exponentially growing negative terms. Proving $\lambda_n \geq 0$ for all n is therefore equivalent to RH [3]. Five concrete open problems are formulated in Part IV.

2.4 Connection to Connes' Program

Part IV additionally provides numerical evidence for the open problem identified in Connes' 2025 survey [2]: that the smallest eigenvalue of the Weil quadratic form QW_λ is simple with even eigenfunction (Theorem 6.1). Using the corrected archimedean kernel $e^{u/2}/(2\sinh u)$ and Cauchy interlacing in a cosine Galerkin basis, we verify that the even-sector spectral gap is strictly positive for all $\lambda \in [5, 200]$. The even–odd sector comparison confirms even dominance for $\lambda \geq 30$, with the crossover from odd to even ground state at $\lambda \approx 28$.

3 Final Status

The FST-RH program provides a thermodynamic and gauge-theoretic perspective on the Riemann Hypothesis, identifying several structural features (archimedean dominance, spectral census, boundary-layer positivity) and producing the strongest known numerical evidence for Connes' Theorem 6.1 prerequisites. It does *not* constitute a proof or a reduction to a simpler problem. The framework's value lies in its conceptual organization, the identification of precise analytical obstructions, and the numerical verification of spectral conditions for the Weil quadratic form.

References

[1] Enrico Bombieri and Jeffrey C. Lagarias. Complements to Li's criterion for the Riemann hypothesis. *Journal of Number Theory*, 77(2):274–287, 1999.

[2] Alain Connes. Letter to Riemann. *arXiv preprint*, 2025. Survey on the Riemann Hypothesis via the Weil quadratic form and prolate spheroidal wave functions.

[3] Xian-Jin Li. The positivity of a sequence of numbers and the Riemann hypothesis. *Journal of Number Theory*, 65(2):325–333, 1997.